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SafeFactory BD

An **ISO 45001-aligned HSE management and site risk intelligence system** for Bangladeshi food and dairy manufacturing — a working management system, not a dashboard, with an analytics layer built on open national data.

36	35	1.05M	2,123	87
HAZARDS ASSESSED	COMPLIANCE REQUIREMENTS	HOURLY AIR RECORDS	GEOCODED FACILITIES	UNIT TESTS PASSING

WHAT IT DOES

HIRA / JSA engine	5×5 matrix with a fatality-escalation rule and hierarchy-of-controls scoring; a 36-hazard dairy and snack-plant register covering ammonia refrigeration, milk-powder dust, hot-oil frying, gas ovens, boilers and confined spaces — every hazard citing its statute.
Permit to Work	Seven permit classes with validity, gas-test, standby and fire-watch rules; automatically detects expired, self-issued, ungas-tested and unclosed permits.
Lockout–Tagout	Equipment-specific isolation sheets across eight energy types; refuses sign-off on an electrical-only isolation of plant holding stored pressure, gravity or thermal energy.
Incidents	Statutory classification; LTIFR, TRIR, DART and severity rate each on its own correctly-named base; Pareto, reporting-health check, 5-Why with a blame detector, bow-tie with barrier-gap detection.
Compliance matrix	35 requirements mapping every ISO 45001:2018 clause group to the Bangladeshi instrument that makes it a legal duty — Labour Act 2006, Labour Rules 2015, BNBC 2020, Fire Prevention and Extinction Act 2003, ECA 1995 — with named evidence and weighted scoring (15 rated Critical).
Site Risk Index	A 0–100 screening score for industrial locations from ambient air burden, industrial density, worker concentration and emergency access — with weight and radius sensitivity analysis.

THREE FINDINGS THAT CAME OUT OF BUILDING IT

1. Separating what a plant *has* from what it *plans* changes the picture.

A standard register reports residual risk, which silently assumes every listed control is installed. Crediting only the controls that actually exist, **26 of 36 hazards sit at High or Critical today** — against 0 in the target state and 32 with no controls at all. The gap is the programme: 31 hazards worth 246 risk points, ranked by the points each outstanding control would remove rather than by raw score.

2. The pipeline refuses to publish a finding it cannot support.

The source air-quality dataset yields a striking Dhaka trend of +2.55 µg/m³ per year over 26 years. It is an artefact: the 2000–2021 segment rises monotonically *every single year* (monotone fraction 1.00, R^2 0.994) then steps down by 68 µg/m³. An automated integrity screen now detects this and **rejects the trend**. The same screen found that **3 pairs of the 30 named cities carry byte-identical hourly series** — two of them over 30 km apart — leaving 27 independent locations, not 30. Both defects would have corrupted every downstream figure and neither is visible in a summary statistic.

3. The exposure analysis ends in a control, not a chart.

Mean PM2.5 at the Dhaka reference point is 48.7 µg/m³, 9.7× the WHO annual guideline, with 58% of hours at ‘Unhealthy for Sensitive Groups’ or worse. An outdoor-work protocol triggered at AQI 150 would fire on **435 of 1,207 valid days (36%)**, concentrated in a predictable dry season (98 µg/m³ in January against 14 in July, a 6.8-fold swing) — so seasonal scheduling is a workable control. Had the figure come out at 80% of days, the honest conclusion would have been the opposite: that a day-by-day trigger is theatre and the budget belongs in engineering controls.

APPLIED TO TWO OPERATING SITES

Site	Index	Plants ≤10 km	What the score means for the HSE plan
Factory 2 — Gazipur	**67.5** Elevated	340	Exposure is **external**: ~278,412 workers inside the same response envelope. Needs mutual-aid arrangements, a joint evacuation understanding with immediate neighbours, and a kept-clear fire-service access route.
Factory 1 — Vulta, Rupganj	**48.7** Moderate	48	Exposure is **internal**: 13 km from a significant urban centre, so help is further away. Needs stronger on-site fire and first-aid capability and a written arrangement with a burn-capable hospital.

Repository github.com//safefactory-bd · **Built with** Python, pandas, Streamlit and Plotly. 87 unit tests, every rate checked against a hand-worked calculation. Eight-page dashboard; one-command launch. Full technical report and source in the repository. **Scope, stated plainly:** the demonstration incident register is synthetic and labelled as such throughout; the hazard register is built from published operations and standard dairy and snack-processing hazard profiles, not from a site walk-through; site coordinates are geocoded to sub-district level, not surveyed; the air-quality data is a gridded model product, not station measurements. This is a decision-support tool, not a substitute for a competent person's assessment of an actual workplace.